Leave Request & Approval – Project Write-up

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Demo Video Link

1. Business problem

Employees need a simple way to request leave, and managers need a way to approve it while keeping balances accurate and seeing who is away. The MVP covers submitting and cancelling requests, a two-step approval, automatic balance updates, a calendar record of approved leave, and a report showing leave by month and team with remaining balances.

2. Solution overview

Everything was built inside one Dataverse solution called **Leave Management** (publisher prefix `lm`) in a Power Apps developer environment.

Layer	Tool	What it does
Data	Dataverse	Leave Request, Leave Balance and Reason Training tables, choice columns, business rules
App	Power Apps (canvas)	Submit, cancel and approve leave, with validation
Automation	Power Automate	Balance update and calendar record on approval, AI tagging of the reason text
Analytics	Power BI	Heatmap by month and team, remaining balance, row-level security
AI	AI Builder	Custom category classification model for the reason text

3. Data model

Leave Request: Employee (lookup to User), Leave Type (choice: Annual, Sick, Other), Start Date and End Date (date only), Reason (multi-line text), Status (choice: Draft, Pending Manager, Pending HR, Approved, Rejected, Cancelled; default Draft), Total Days (formula column), Manager and HR Approver (lookups to User), AI Category (text).

Leave Balance: Employee, Leave Type, Entitled Days, Used Days, and Remaining Days (formula column: Entitled minus Used).

Business rules

- On Leave Request, an error message appears if End Date is earlier than Start Date.
- On Leave Balance, Used Days defaults to 0 when it has no value.

4. Canvas app

The app has three screens.

- **SubmitRequest**: leave type, dates, reason, and a live total-days label. The Submit button checks that the end date is not before the start date, warns if the dates overlap an existing active request, and blocks the request if the days exceed the remaining balance. A valid request is saved with the status Pending Manager.
- **MyRequests**: a gallery of the signed-in user's requests with type, dates and status. A Cancel button is shown only while a request is pending.
- **ApproverScreen**: a gallery of requests waiting at Pending Manager or Pending HR, with checkboxes and bulk **Approve selected** and **Reject selected** buttons. Approving a Pending Manager request moves it to Pending HR, and approving a Pending HR request marks it Approved.

5. Automation (Power Automate)

Flow 1: Leave Approved – Update Balance and Calendar. It triggers when a request's Status changes. If the new status is Approved, the flow:

1. Calculates the number of leave days from the start and end dates.
2. Finds the employee's Leave Balance rows and picks the one matching the leave type.
3. Adds the days to Used Days, which lowers Remaining Days automatically.
4. Creates an all-day Appointment row in Dataverse for the leave period.

Flow 2: Tag Leave Reason with AI. When a new request is added and has a reason, the flow sends the reason text to the AI Builder model and saves the predicted category in AI Category.

6. Analytics (Power BI)

The report uses Import mode with these tables: Leave Request, Leave Balance, an Employees table (employee, team, manager email), a date table, and a LeaveDays table that expands each approved request into one row per leave day so month totals are correct.

- A **matrix heatmap** of approved leave days by team and month (with colour gradient).
- A **bar chart** and a **card** for used versus remaining days by leave type, plus a leave type slicer.
- **Row-level security:** a role called Manager filters the Employees table with `[ManagerEmail] = USERPRINCIPALNAME()`. Because Employees filters all other tables, each manager sees only their own team. The role was tested in Power BI Desktop (View as) with a valid manager email, which showed data, and with an unknown email, which showed an empty report. The report was published to the Power BI Service and the role was assigned there.

7. AI Builder stretch goal

A custom **category classification** model was trained on 60 labelled reason texts (20 each for Annual, Sick and Other) and published. In testing, "Fever and cold, need rest" was classified as Sick with a confidence of about 0.94, and "Going for a trip" was classified as Annual. The AI category is stored beside the leave type the employee chose, so a manager can spot a mismatch.

8. Acceptance criteria

Criterion	Result
Two-step approval works	Met. Status moves Pending Manager, then Pending HR, then Approved.
Balances decrement	Met. Used Days increases and Remaining Days falls after final approval.
Calendar shows approved leave	Partly met. Approved leave is written to the Dataverse Appointment table (see limitations).
Conflict warning on overlap	Met for overlapping date ranges (see limitations).
Heatmap and remaining balance	Met.
Row-level security	Met.
Stretch: category classification	Met.

9. Limitations and assumptions

- **Outlook invites:** the Outlook connector failed with a mailbox error because the account has no Exchange Online licence. Approved leave is stored as a Dataverse Appointment instead. With a mailbox, the Create event (V4) step can be added to the same flow.
- **Single test user:** testing used one account acting as employee, manager and HR approver. The Employees table in Power BI is entered by hand, so teams such as HR and Finance appear as sample rows without leave data.
- **Approver filtering:** the approver screen shows all requests at the pending stages. Filtering by the assigned Manager and HR Approver is a planned improvement.
- **Overlap on identical dates:** the overlap check compares dates in a way that can be affected by time zone offsets. A fix that adjusts the picked dates by the UTC offset is documented, and same-day cases should be retested.
- **Power BI refresh:** the published report uses imported data and needs a refresh or republish to show new Dataverse data.

10. Next steps

1. Connect Exchange Online and add the Outlook calendar invite to the approval flow.
2. Filter the approver screen by the assigned approver, and look up each employee's manager automatically.
3. Replace the manual Employees table with a Team column in Dataverse.
4. Add scheduled refresh in the Power BI Service.
5. Notify employees by email or Teams when a request is approved or rejected.

11. Screenshots:

Tables:

Leave Balance

Leave Balance columns and data

Name	Used Days	Modified On	Remainin...	Entitled ... *	+19 more
Niya - Annual	11	10/29/2026 2:33 PM	19.00	30	
Niya - Other	0	10/22/2026 6:35 PM	5.00	5	
Niya - Sick	12	10/29/2026 2:44 PM	8.00	20	
Enter text	*Enter number*			*Enter number*	

Leave Request

Leave Request columns and data — Update forms and views — Edit

Name	End Date *	Start Date *	Status *	AI category	Created By	Created On	Leave Type *	+21 more
# niyu - Annual	9/30/2026	9/25/2026	Pending Man...		# niyu	10/22/2026 8:16 PM	Annual	
# niyu - Annual	9/25/2026	9/25/2026	Cancelled		# niyu	10/22/2026 8:17 PM	Annual	
# niyu - Annual	11/28/2026	11/26/2026	Rejected		# niyu	10/23/2026 8:44 AM	Annual	
# niyu - Annual	12/19/2026	12/17/2026	Cancelled		# niyu	10/23/2026 8:52 AM	Annual	
# niyu - Annual	1/6/2027	1/4/2027	Approved		# niyu	10/23/2026 1:00 PM	Annual	
# niyu - Annual	1/8/2027	1/7/2027	Approved		# niyu	10/23/2026 1:17 PM	Annual	
# niyu - Annual	1/12/2027	1/11/2027	Approved		# niyu	10/23/2026 1:31 PM	Annual	
# niyu - Annual	2/11/2027	2/10/2027	Approved	Annual	# niyu	10/29/2026 1:30 PM	Annual	
# niyu - Annual	2/13/2027	2/13/2027	Approved	Sick	# niyu	10/29/2026 2:04 PM	Annual	
# niyu - Annual	2/16/2027	2/15/2027	Approved	Annual	# niyu	10/29/2026 2:08 PM	Annual	

+ 11 additional rows

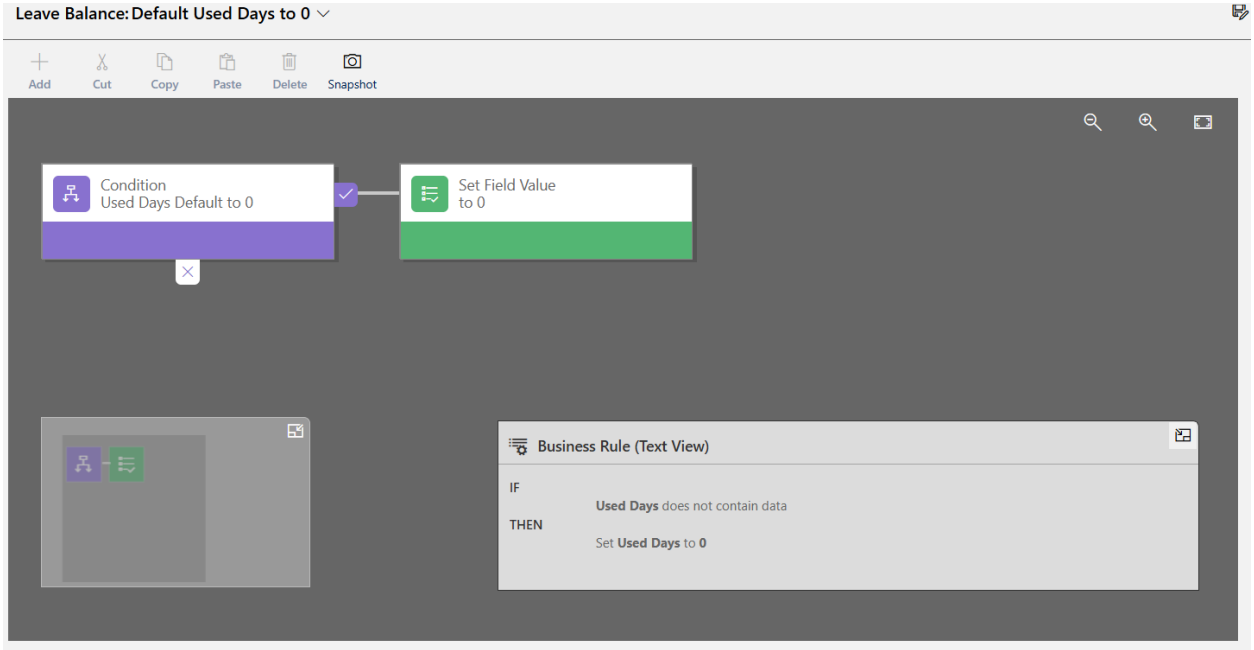
Reason Training

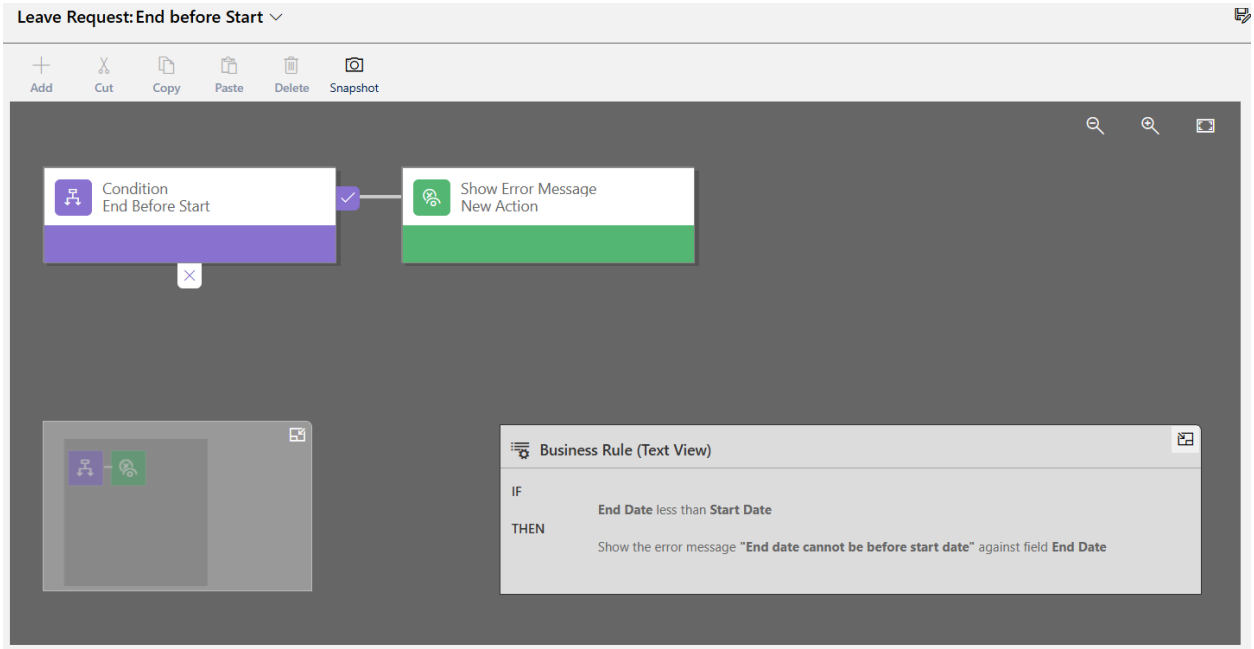
Reason Training columns and data

New column	ReasonText *	+20 more
Annual	Family vacation to Goa	
Annual	Trip with friends to the hills	
Annual	Holiday with family	
Annual	Travelling abroad for a break	
Annual	Planned vacation at the beach	
Annual	Going on a long weekend getaway	
Annual	Need a rest and some time off	
Annual	Visiting my hometown for a holiday	
Annual	Honeymoon trip	
Annual	Summer break with kids	

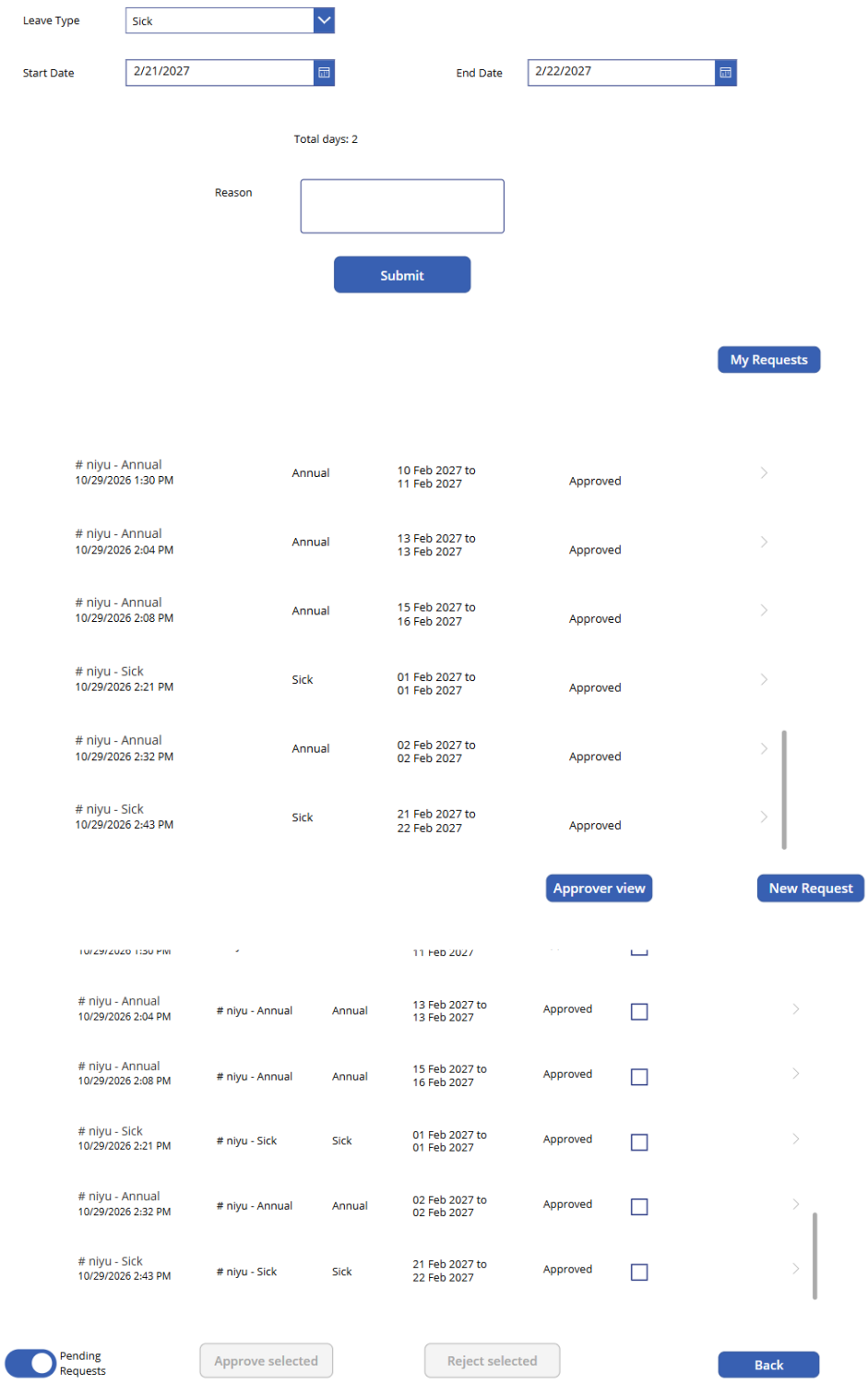
+ 50 additional rows

Business Rules:

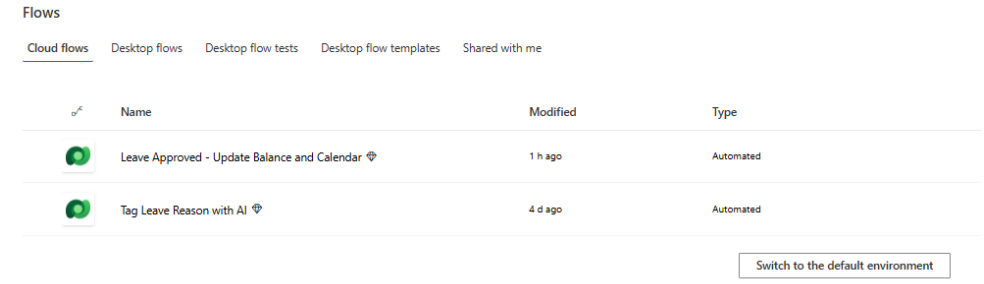


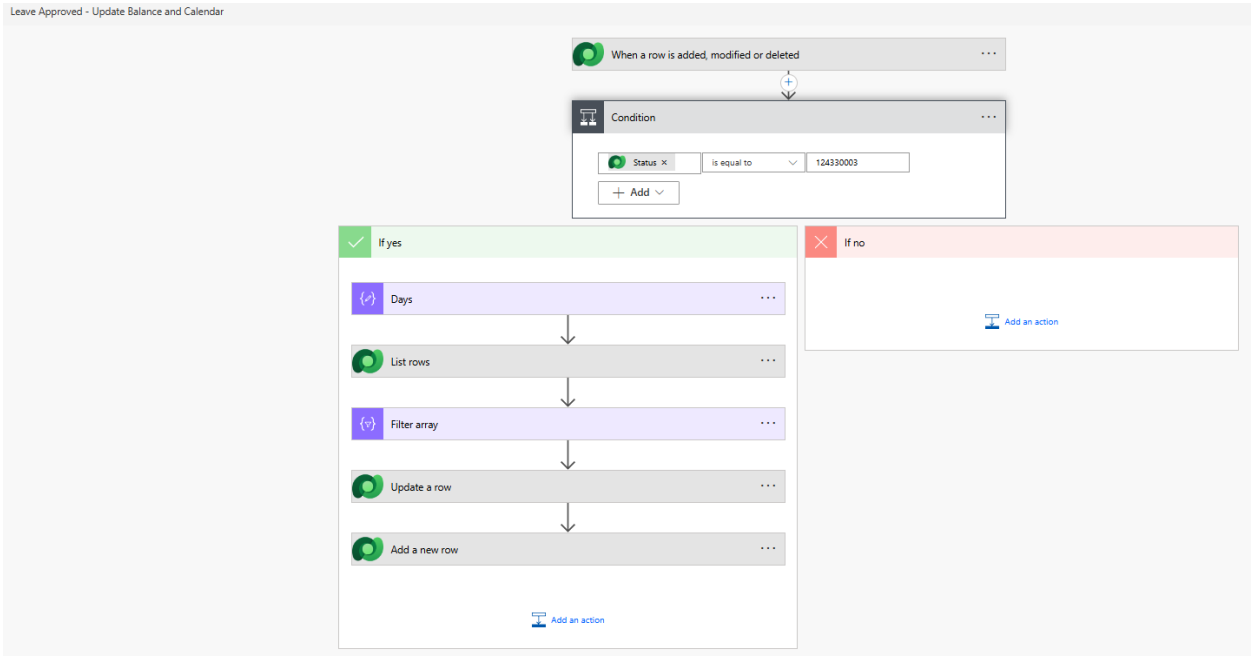


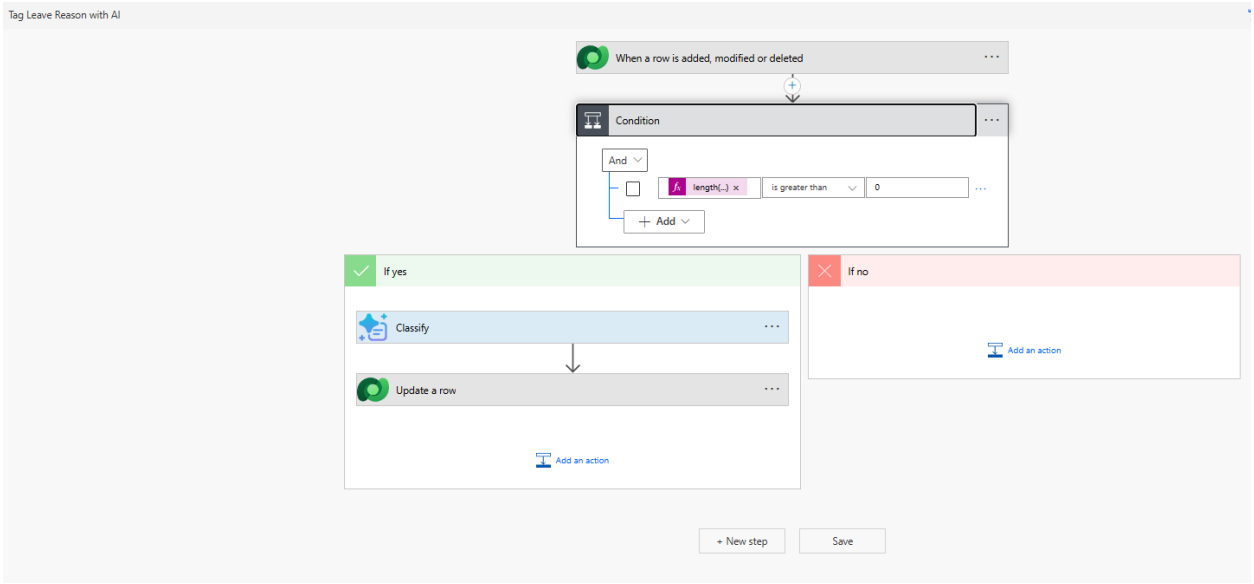
Leave App:


Leave Type
Sick
Start Date
2/21/2027
End Date
2/22/2027
Total days: 2
Reason
Submit
My Requests
niyu - Annual
10/29/2026 1:30 PM
Annual
10 Feb 2027 to
11 Feb 2027
Approved
niyu - Annual
10/29/2026 2:04 PM
Annual
13 Feb 2027 to
13 Feb 2027
Approved
niyu - Annual
10/29/2026 2:08 PM
Annual
15 Feb 2027 to
16 Feb 2027
Approved
niyu - Sick
10/29/2026 2:21 PM
Sick
01 Feb 2027 to
01 Feb 2027
Approved
niyu - Annual
10/29/2026 2:32 PM
Annual
02 Feb 2027 to
02 Feb 2027
Approved
niyu - Sick
10/29/2026 2:43 PM
Sick
21 Feb 2027 to
22 Feb 2027
Approved
Approver view
New Request
niyu - Annual
10/29/2026 2:04 PM
niyu - Annual
Annual
13 Feb 2027 to
13 Feb 2027
Approved
niyu - Annual
10/29/2026 2:08 PM
niyu - Annual
Annual
15 Feb 2027 to
16 Feb 2027
Approved
niyu - Sick
10/29/2026 2:21 PM
niyu - Sick
Sick
01 Feb 2027 to
01 Feb 2027
Approved
niyu - Annual
10/29/2026 2:32 PM
niyu - Annual
Annual
02 Feb 2027 to
02 Feb 2027
Approved
niyu - Sick
10/29/2026 2:43 PM
niyu - Sick
Sick
21 Feb 2027 to
22 Feb 2027
Approved
Pending
Requests
Approve selected
Reject selected
Back

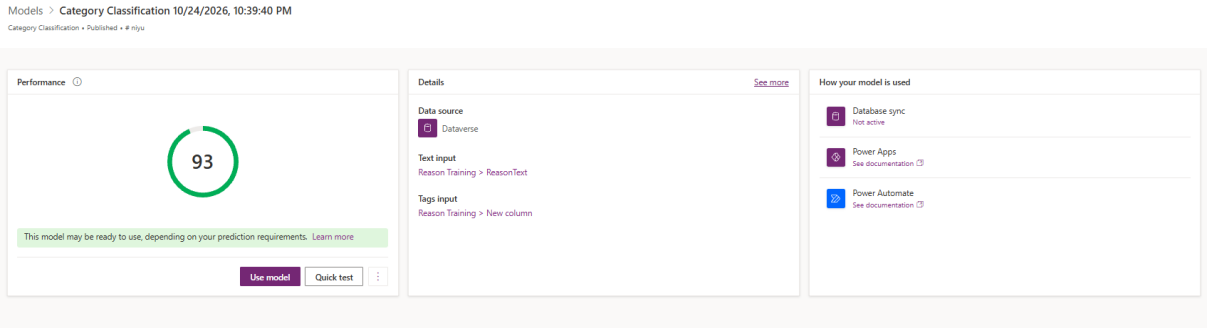
Cloud Flows:


Flows
Cloud flows
Desktop flows
Desktop flow tests
Desktop flow templates
Shared with me
Name
Modified
Type
Leave Approved - Update Balance and Calendar
1 h ago
Automated
Tag Leave Reason with AI
4 d ago
Automated
Switch to the default environment


Leave Approved - Update Balance and Calendar
When a row is added, modified or deleted
Condition
Status
is equal to
124330003
Add
If yes
If no
Days
List rows
Filter array
Update a row
Add a new row
Add an action
Add an action


Tag Leave Reason with AI
When a row is added, modified or deleted
Condition
And
length(...)
is greater than
0
Add
If yes
If no
Classify
Update a row
Add an action
Add an action
+ New step
Save

AI Builder:



Power BI Report:

